

数据库系统 第三次作业

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3.8

(a) (SELECT ID FROM depositor) EXCEPT (SELECT ID FROM borrower);

(b) SELECT F.ID FROM customer AS F, customer AS S WHERE
F.customer_street = S.customer_street AND F.customer_city = S.customer_city
AND S.customer_id = '12345';

(c) SELECT DISTINCT branch_name FROM account, depositor, customer
WHERE customer_id = depositor.id AND depositor.account_number = account.account_
number AND customer.customer_city = 'Harrison'.

3.9 (a) SELECT e.ID, e.person_name, e.city FROM employee AS e, works AS w
WHERE w.ID = e.ID AND w.company_name = 'First Bank Corporation'.

(b) SELECT e.ID, e.person_name, e.city FROM employee AS e, works AS w WHERE
w.ID = e.ID AND w.company_name = 'First Bank Corporation' AND w.salary > 10000.

(c) SELECT ID FROM works WHERE company_name <> 'First Bank Corporation'.

(d) SELECT ID FROM works WHERE salary > ALL(SELECT salary FROM works WHERE
company_name = 'Small Bank Corporation'). 见反面.

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(e) SELECT S.company_name FROM company AS S WHERE NOT EXISTS ((SELECT city FROM company WHERE company_name = 'Small Bank Corporation') EXCEPT (SELECT city FROM company AS T WHERE S.company_name = T.company_name)).

(f) SELECT company_name FROM works GROUP BY company_name HAVING COUNT(DISTINCT ID) >= ALL (SELECT COUNT(DISTINCT ID) FROM works GROUP BY company_name).

(g) SELECT company_name FROM works GROUP BY company_name HAVING AVG(SALARY) > (SELECT AVG(salary) FROM works WHERE company_name = 'First Bank Corporation').

3.10 (a) UPDATE employee SET city = 'Newtown' WHERE ID = '12345'

(b) UPDATE works T SET T.salary = T.salary * 1.03 WHERE T.ID IN (SELECT manager_id FROM manages) AND T.salary * 1.1 > 100000 AND T.company_name = 'First Bank Corporation';

UPDATE works T SET T.salary = T.salary * 1.1 WHERE T.ID IN (SELECT manager_id FROM manages) AND T.salary * 1.1 <= 100000 AND T.company_name = 'First Bank Corporation'.

3.15 (a) SELECT c.ID, c.customer_name FROM customer c JOIN depositor d ON c.ID = d.ID JOIN account a ON d.account_number = a.account_number WHERE a.branch_name IN (SELECT branch_name FROM branch WHERE branch_city = 'Brooklyn') GROUP BY c.ID, c.customer_name HAVING COUNT(DISTINCT a.branch_name) = (SELECT COUNT(*) FROM branch WHERE branch_city = 'Brooklyn'); (b) SELECT SUM(amount) FROM loan.

(c) SELECT branch_name FROM branch WHERE assets > SOME (SELECT assets FROM branch WHERE branch_city = 'Brooklyn');